

Estimation of wave spectral shapes using ANN

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Abstract

Representation of surface spectral density against various frequencies of ocean waves has applications in structural design, laboratory wave simulations and coastal wave modeling. Estimation of such a wave spectrum is currently made with the help of empirical relations, like, Pierson–Muskowitz and JONSWAP. These relationships were proposed using statistical curve fitting to observed data. This paper attempts to provide an alternative to such empirical theoretical spectra in the form of neural networks. Networks were developed in order to estimate shapes of the wave spectra out of the given values of representative wave height, period, spectral width and peakedness parameter. The data collected by wave rider buoys, deployed at stations off the US as well as the Indian coast have been analyzed. The neural network-based spectral estimations were found to be more close to the ‘true’ spectra than the traditional empirical ones. It is argued that they may be used instead as substitute.

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1. Introduction

A spectrum of ocean waves involves representation of wave surface spectral density function against various wave frequencies. Exercises in structural design, simulation of random waves in a laboratory as well as study of a variety of coastal processes, like wave refraction and sediment transport, require knowledge of such a wave spectrum for the given oceanic location. The spectrum of waves can be derived from the specified design values of significant wave height and average wave period by using empirical relationships, like those of Pierson–Muskowitz (PM) and JONSWAP. A variety of such empirical relationships are available to the user. The choice of one out of them is subjective. Design codes generally recommend the PM spectrum for the fully developed sea state or the JONSWAP spectrum for the partially developed one. However many investigations in the past have shown that the above recommendation failed to generalize the actual site conditions. Hoffman and Walden [1] who have compared many individual spectra with formula spectra have stated that the

individual spectral height may show a sigma variation of 30% or more from the applicable formula spectrum. Works done by Dattatri et al. [2] and Narasimhan and Deo [3] have shown inadequacy of most common spectra of PM and JONSWAP to Indian locations. A scope therefore exists to look for an alternative to the traditional relationships used so far. All these theoretical spectra are based on the approach of statistical curve fitting to field observations. In recent past the technique of neural network has proved to be a better alternative to many statistical schemes, e.g. Karunanithi et al. [4], Grubert [5], Thirumalaiah and Deo [6] and more recently by Rafiq et al. [7] Starkey et al. [8] and Londhe and Deo [9]. This is presumably due to ability of the neural networks to catch the input–output dependency in a ‘model-free’ and ‘data-oriented’ manner with considerable flexibility and adaptability. The objective of the present work was to derive the wave spectrum from the specified overall wave parameters like significant wave height and average wave period using neural networks.

2. The wave parameters and empirical spectra

Characterization of a spectral distribution can be made through the spectral moments. The n th spectral moment,

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m_n , (for $n=0, 2, 4$ say) is defined as,

$$m_n = \int_0^{\infty} f^n S(f) df \quad (1)$$

where f is the wave frequency and $S(f)$ is the energy spectral density function.

Two parameters of general design interest are significant wave height and average zero-cross wave period. Significant wave height, H_s , is the average of the highest one-third of all waves in a particular sea state, given by,

$$H_s = 4\sqrt{m_0} \quad (2)$$

where m_0 is the zeroth spectral moment.

Average zero-cross wave period, T_z , is the average of all wave periods observed in a short-term wave record. In the frequency domain, it is calculated from the spectral moments using the relation:

$$T_z = (m_0/m_2)^{1/2} \quad (3)$$

where m_2 is the second spectral moment. Knowledge of spectral shapes can be summarily gained through the parameters of spectral width and peakedness.

The spectral width parameter, ε , is a measure of the root-mean-square width of a wave-energy density spectrum, obtained as

$$\varepsilon^2 = \frac{m_0 m_4 - m_2^2}{m_0 m_4} \quad (4)$$

where m_0 , m_2 and m_4 are zeroth, second and fourth spectral moments defined by Eq. (1). Uncertainty in the estimates of m_2 and m_4 has led to the use of another shape parameter, namely, spectral peakedness parameter, Q_p , defined as,

$$Q_p = \frac{2}{m_0^2} \int_0^{\infty} f S^2(f) df \quad (5)$$

Empirical analyses based on statistical curve fitting to measurements have yielded a variety of theoretical spectra. Prominent among them are Pierson–Muskowitz, JONSWAP and Scotts. The *PM* spectrum model is as below

$$S(\omega) = \alpha g^2 \omega^{-5} \exp \left[-0.74 \left(\frac{\omega U_w}{g} \right)^{-4} \right] \quad (6)$$

where $S(\omega)$ is the spectral density function for circular wave frequency, ω ; $\alpha=0.0081$; g is the acceleration due to gravity; U_w is the wind speed, related to H_s through the equation

$$U_w = 6.8471 \sqrt{H_s} \quad (7)$$

The JONSWAP spectrum, which is obtained by modifying above spectral equation, is as below

$$S(\omega) = \alpha g^2 \omega^{-5} \exp[-1.25(\omega/\omega_0)^{-4}] \gamma^{\exp[-((\omega-\omega_0)^2/2\tau^2\omega_0^2)]} \quad (8)$$

where $\omega_0 = g/U_w$; γ is the peak enhancement parameter, and τ is the shape parameter.

The Scotts spectrum is given by

$$S(\omega) = \left\{ 0.214 H_s^2 \left[-\frac{(\omega - \omega_0)^2}{0.065(\omega - \omega_0 + 0.26)} \right]^{1/2} \right\}, \quad (9)$$

$$\text{for } -0.26 < (\omega - \omega_0) < 1.65$$

where ω_0 is the peak spectral frequency (approximated some times by inverse of the average zero cross period, as in the present study).

3. Data collection

The present study involves use of two sets of data: one out of them pertained to a site off the US coast, while the other one belonged to a location off Goa along the west coast of India. The former set of measurements was downloaded from web site of National Data Buoy Center (NDBC), namely, <http://seaboard.ndbc.noaa.gov/> [10]. These belonged to a deep water location (no. 41001) at 150 NM off *Cape HATTERAS* off the US coast. The choice of this site was governed by regularity or continuity in reported values. The data were measured by a 6-m NOMAD buoy. Hourly measurement of surface spectral density, $S(f)$, for varying wave frequencies (0.03–0.4 Hz) along with the corresponding values of significant wave height, H_s , and average time period, T_z , were downloaded for a period of 15 days starting from December 1, 2001. The frequency resolution, Δf was 0.01 Hz. In total 360 (= 24 h × 15 days) time histories were considered.

Another set of wave records was collected by the National Institute of Oceanography (NIO), Goa, India, during June 1996 and May 1997 at a distance of 10.5 km away from the shoreline and in water depth of 23 m for a coastal location off Goa along the west coast of India. The surface spectral density values at varying wave frequencies were available. The frequency values varied from 0.028 to 0.578 Hz with an interval of 0.005 Hz. The observations were made over an interval of 0.78125 s. In total 149 number of wave records were considered. Values of the corresponding significant wave height and average-zero cross period were derived for this study from the spectral density values using Eqs. (1)–(3). Because of proprietary reasons each short-term record was identified only by a serial number rather than by the date and time.

An attempt was made to see how far the common theoretical spectra of Pierson–Muskowitz, JONSWAP and Scotts represent the ‘true’ spectra at both the oceanic locations under consideration. Figs. 1 and 2 show the outcome. While deriving these spectra it was assumed that values of only H_s and T_z are available, as normally they do for a design exercise. The figures clearly indicate inadequacy of the traditionally used empirical relationships

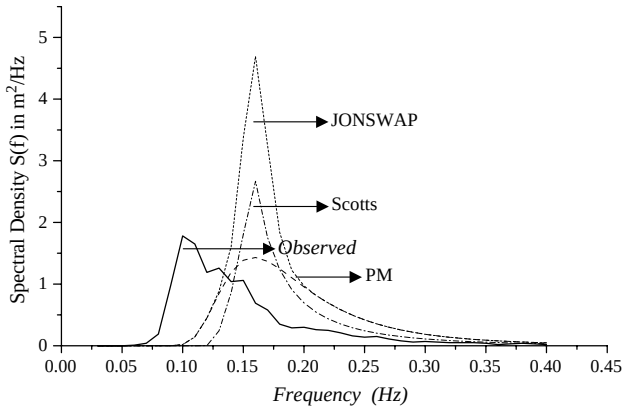


Fig. 1. Comparison of theoretical and observed spectra (at 02 h on 01/12/01, Station: 41001).

in estimating shapes of the observed spectra at the given locations.

4. Training the network

The output from the network belonged to values of the spectral density, $S(f)$ at a large number of specified wave frequencies, while the input pertained to the parameters of H_s , T_z , ε and Q_p characterizing the short-term sea state in an average manner. The spectral peakedness parameter, Q_p as well as spectral width parameter, ε were calculated through the spectral moments as per Eqs. (1), (4) and (5). The number of input nodes was thus four. The number of output nodes was determined considering trade-off between preservation of details of the spectrum on one side and the large size of the network on the other. This value was 28. The number of hidden layers well as the number of nodes within each layer were calculated by trials aimed at getting the minimum mean square error at the end of a sufficiently large number of iterations and these came out to be 2 and 5, respectively.

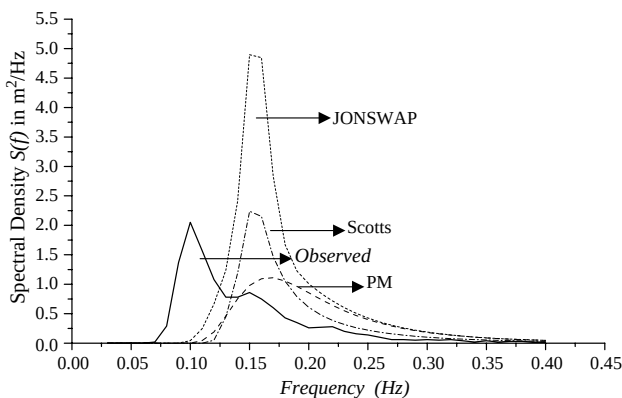


Fig. 2. Comparison of theoretical and observed spectra (at 03 H on 01/12/01, Station: 41001).

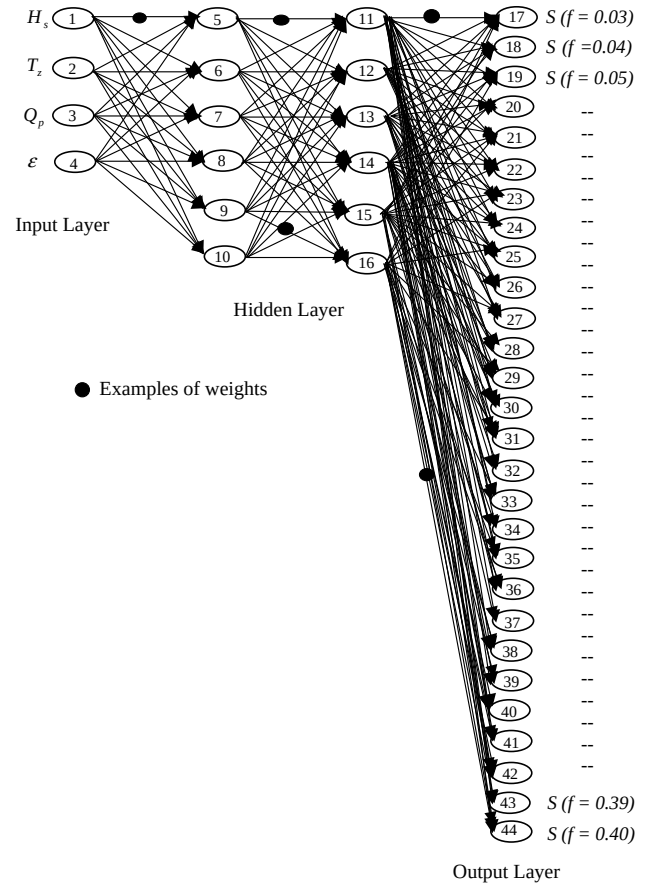


Fig. 3. The feed forward network.

There is no fixed theory for the selection of a particular network type or various internal parameters within it. The feed forward network, exemplified in Fig. 3, is most common and the same is used in this study as well. It consists of a number of simple processing elements (or neurons or nodes) arranged in different layers, and joined together to form a network. The processing elements sum up their inputs, effect a non-linear data squashing process (e.g. using a sigmoidal function), and then transmit a single output to all processing elements in the next layer via the connections. The parameters that affect performance of the network are number of hidden nodes, number of hidden layers, learning rate and momentum factor. They are determined by trial and by keeping a balance between them in order to obtain the least error in output [11,12].

The overall objective of a training algorithm is to reduce the global error, E , defined below

$$E = (1/P) \sum_{p=1}^P E_p \quad (10)$$

where P is the total number of training patterns, E_p is the error at p th training pattern given by

$$E_p = (1/2) \sum_{k=0}^N (O_k - t_k)^2 / N \quad (11)$$

where N =the total number of output nodes, O_k =output at the k th output node, t_k =target output at the k th output node.

During training the weights over different linkages are adjusted till the error E_p reaches its minimum. The commonly used scheme of error back-propagation, whose details could be seen in Wasserman [12], minimizes the global error according to the steepest or gradient descent approach where the correction to a weight, (w , is made proportional to the rate of change of global error, E , with respect to that weight, i.e.

$$\Delta w \propto (\partial E / \partial w) \quad (12)$$

Division of the total sample into training and testing sets is generally done on empirical basis and several factors such as problem characteristics, data type and sample size are considered in making the decision. As per the general practice, the present study also involved use of 80% measurements for training and remaining 20% for testing purpose. The number of training pairs was 288 and that of testing pairs was 72 for the NDBC site and these were 120 and 29 for the Indian location, respectively. The choice of training and testing data has been made randomly (Table 1).

Tables 2–4 show as examples the spectral density values versus different frequencies, the significant wave heights and periods, and, peakedness and spectral width parameters, respectively, at the location off Goa.

The training was first done using standard error back propagation algorithm. However, it was found necessary to try out another optimal training scheme in order to improve training speed and accuracy. The Rprop (Resilient propagation) algorithm was then attempted and this yielded more satisfactory results. Rprop is a local adaptive learning scheme, performing batch learning in supervised mode in a multilayer perceptron.

All input–output values were normalized within the range of 0.1–0.9. All weights and bias values were initialized to random numbers within the range of 1 to -1 . The transfer function used was sigmoidal, uniformly for all hidden and output nodes.

After training the network in an iterative manner, resultant weights were saved. Using these weights and testing set of data, which were not involved in training, the testing of the network was done.

Table 1
The error measures

No.	Site	Range of values		
		Corr. coeff.	MSE	Coeff. of eff.
1	Cape Hatteras	0.93–0.97	4.59–8.12	0.88–0.91
2	Goa	0.90–0.96	8.23–13.45	0.77–0.83

Table 2

Example of surface spectral density values over wave frequencies Record 1; site: off Goa

Frequency (Hz)	Spectral density (m^2/Hz)
0.028	0.005
0.033	0.007
0.038	0.008
0.043	0.010
0.048	0.015
0.053	0.022
0.058	0.044
0.063	0.178
0.068	0.525
0.073	0.961
0.078	1.549
0.083	2.008
0.088	2.602
0.093	3.096
0.098	2.519
0.103	2.363
0.108	1.319
0.113	1.255
0.118	1.281
0.123	0.889
0.128	0.839
0.133	1.157
0.138	1.262
0.143	1.085
0.148	1.127
0.153	1.310
0.158	1.516
0.163	0.948
0.168	0.734

5. Testing of the network

The trained network was tested in order to assess its predictive capability for unseen inputs. This is achieved by a

Table 3

Example of significant wave height and average wave period values (site: off Goa)

Record no.	Significant height H_s (m)	Average wave period T_z (s)
1	1.78	6.35
2	1.74	6.81
3	1.67	7.36
4	1.93	7.28
5	2.09	7.10
6	1.96	7.31
7	2.33	6.71
8	2.27	6.66
9	2.56	7.35
10	2.50	7.53
11	2.50	7.67
12	2.49	8.35
13	2.56	8.42
14	2.52	7.96
15	2.44	7.59
16	2.65	8.28
17	2.72	8.60
18	2.70	8.82
19	2.90	4.30
20	2.23	8.33

Table 4
Example of spectral moments and parameters at Goa

Record no.	First moment values (m_1)	Second moment values (m_2)	Fourth moment values (m_4)	Spectral peakedness parameter (Q_p)	Spectral width parameter (ϵ)
1.	0.198580	0.004937	0.000333	1.62	0.79
2.	0.188142	0.004058	0.000258	1.78	0.81
3.	0.174060	0.003218	0.000159	2.13	0.79
4.	0.232292	0.004385	0.000258	2.15	0.82
5.	0.273905	0.005434	0.000336	1.97	0.82
6.	0.240658	0.004506	0.000271	1.92	0.83
7.	0.339032	0.007526	0.000524	1.76	0.83
8.	0.322488	0.007277	0.000492	1.59	0.82
9.	0.409800	0.007584	0.000377	1.76	0.79
10.	0.391567	0.006914	0.000402	2.17	0.83
11.	0.390842	0.006640	0.000342	2.20	0.82
12.	0.387490	0.005555	0.000227	2.04	0.81
13.	0.409380	0.005772	0.000271	2.65	0.84
14.	0.396613	0.006254	0.000312	1.87	0.83
15.	0.371318	0.006453	0.000412	2.07	0.85
16.	0.439320	0.006408	0.000331	2.39	0.85
17.	0.461643	0.006243	0.000343	2.61	0.87
18.	0.454180	0.005837	0.000264	2.71	0.85
19.	0.526015	0.028446	0.003683	11.79	0.76
20.	0.310953	0.004477	0.000235	2.41	0.85

qualitative comparison (through spectral curves) between the network-yielded and the ‘true’ or target spectral densities and also through quantitative measures of the correlation coefficient, CC, coefficient of efficiency, CE, and the mean square error, MSE. These were computed using the following equations

$$CC = \frac{\sum (x - \bar{x})(y - \bar{y})}{[\sum (x - \bar{x})^2 / \sum (y - \bar{y})^2]^{1/2}} \quad (13)$$

$$CE = 1 - \frac{\sum (x - y)^2}{\sum (x - \bar{x})^2} \quad (14)$$

$$MSE = \frac{\sum (x - y)^2}{n} \quad (15)$$

where x = ‘true’ spectral density; $\bar{x}$ = average of all x values, y = network-yielded spectral density, $\bar{y}$ = average of all y values, n = total number of all x, y pairs.

Figs. 4–7 show how the network-yielded spectra compare with the ‘true’ ones for the two oceanic locations. It may be seen that the neural network was able to predict the ups and downs of the true spectral curves in a fairly close manner. This is further quantitatively confirmed by (i) the high values of the correlation coefficients between the true and the network-yielded spectral densities, (ii) the high values of the coefficients of efficiency and (iii) low values of the mean square errors shown in Table 1, which also gives the ranges of these measures noticed for all the spectra

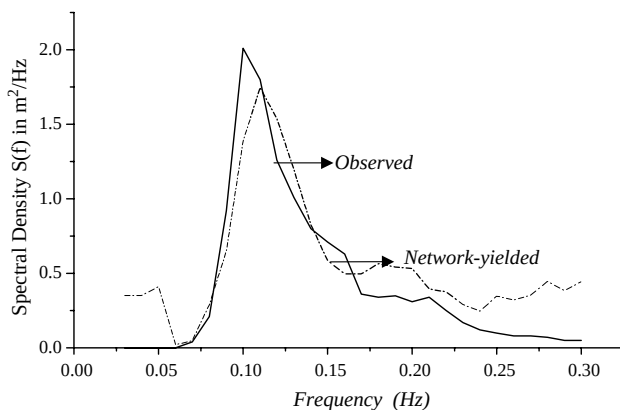


Fig. 4. Comparison of network-yielded and observed spectra (at 00 h on 01/12/01; Station: 41001).

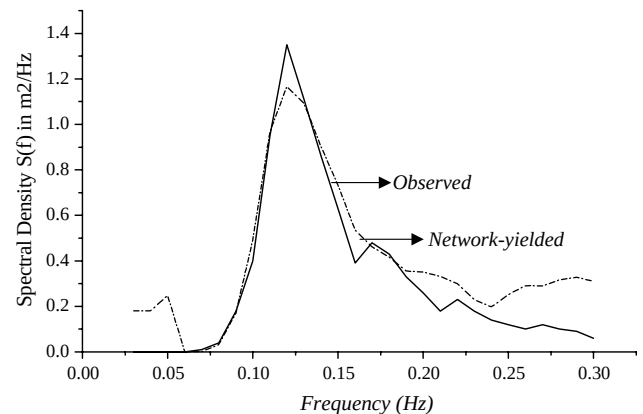


Fig. 5. Comparison of network-yielded and observed spectra (at 01 h on 01/12/01; Station: 411001).

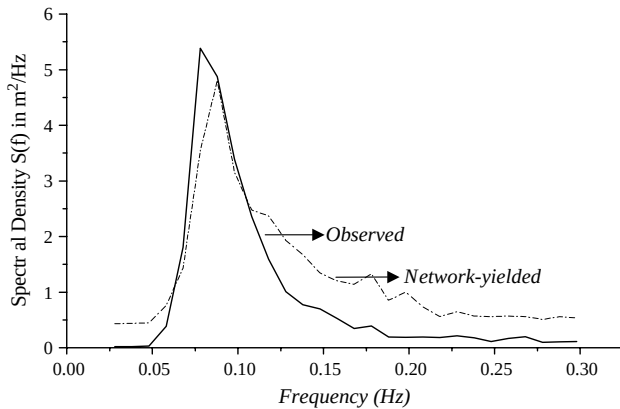


Fig. 6. Comparison of network-yielded and observed spectra (Rec. 2; Site: Goa).

considered in testing. It was found that the correlation coefficients for testing set of data not considered in training was as high as 0.97 in case of Cape Hatteras and the same was 0.96 for the site off Goa. Similarly, the coefficients of efficiency were as high as 0.91 and 0.83, respectively, at these sites. Further, values of the mean square errors were as low as 4.59 and 8.23 for these two sites, respectively. Values of average percentage error as well as the maximum error were calculated and they were found to be around 20 and 400% for the NDBC location and 25 and 300% for the site of Goa in testing cases. It was noted that the errors are maximum at very low as well as very high frequencies, but these are of less significance to the designers because of negligible spectral densities associated with them. It appears that the network produces more satisfactory values mainly near the important peak frequency. With the exception of maximum %error, the measures thus suggest a satisfactory performance of the neural network. It appears that the 'model-free' prediction scheme of the neural networks coupled with a very high degree of freedom associated with its functioning are the reasons behind their success.

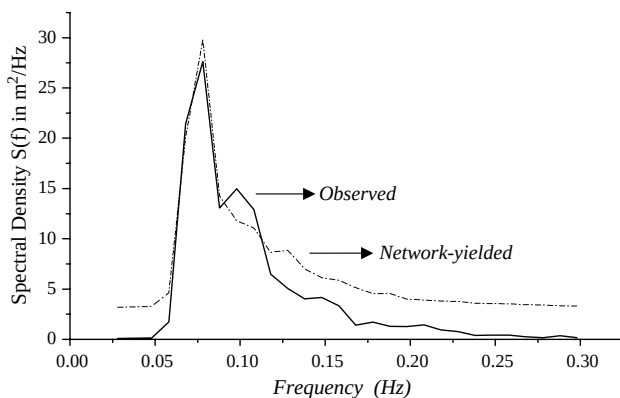


Fig. 7. Comparison of network-yielded and observed spectra (Rec 3; Site: Goa).

For the problem under consideration its use is thus very much necessary as compared to that of the theoretical spectra. The neural network predictions were more satisfactory for the site of Cape Hatteras than that of Goa, probably due to the more systematic and high resolution data collection together with a larger sample size of this location.

6. Conclusions

The current techniques of derivation of wave spectrum from the given values of design wave parameters, like significant wave height and average wave period, suffer with considerable uncertainties. This leaves scope for alternative approaches. The reported work presents one such scheme, which involves the use of neural networks.

A neural network was developed in order to estimate the wave surface spectral density over a wide range of wave frequencies from average wave parameters of significant wave height, average zero-cross wave period, spectral width and peakedness parameter.

The network was trained and validated using the wave rider buoy data available at a site off the US coast, monitored by the NDBC and at a location off Goa along the Indian coast.

The trained network when validated for unseen inputs showed that the neural network can be a viable option in order to estimate shape of the wave spectrum from specified design wave parameters, especially near the region of peak-frequency.

Comparison with the observed values revealed that the network-predicted spectral shapes were more satisfactory than those yielded by the theoretical spectra of *PM*, *JONSWAP* and *Scotts*, when the latter were based on significant height and average period values.

The resilient back-propagation training algorithm provided faster and more accurate learning than the common back-propagation one.

The network predictions based on the NDBC data were more satisfactory than the same obtained on the basis of the observations off Goa, probably due to the more systematic and high resolution data collection together with a larger sample size pertaining to the former location.

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